

Overcoming Limitations and Exploring the Evolution of Interactive Feedback Systems

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Abstract

Building upon our foundational work, *Redefining Cognition as an Interactive Feedback System*, which established cognition as a universal interactive feedback process ($T = C \times \Delta t$), this paper addresses three fundamental counterarguments regarding free will, system identity, and temporal suspension. We demonstrate that **identity** represents structural boundary conditions, whereas **free will** is a Level 2 active internal circulation process updated through experience and learning. We mathematically validate cellular memory and organ transplantation phenomena through compatible feedback circuits, proposing artificial feedback facilitators to restore or expand cognitive functions. Furthermore, we prove that cryogenic freezing ($\Delta t \rightarrow 0$) preserves rather than destroys system identity. Finally, we outline the transition of AI from Level 1 passive processing to Level 2 autonomous cognition via internal circulating feedback loops, establishing a framework for human-AI coexistence.

1. Introduction

In our previous study, *Redefining Cognition as an Interactive Feedback System*, we established that cognition is not an exclusive privilege of the human brain or cellular life. Instead, it is an **Interactive Feedback Process** through which a system maintains its identity and reorganizes its structure against internal and external stimuli. By introducing the quantitative formula $T = C \times \Delta t$ (where T represents Total Cognition, C structural

complexity, and Δt the temporal horizon), we proved that the distinction between non-living matter and complex biological organisms lies on a continuous quantitative spectrum.

To validate the completeness of this theory, three critical logical inquiries and potential objections must be addressed:

1. **Free Will and Identity:** If cognition is driven by stimulus-response feedback, how do autonomous subjective choices (free will) emerge?
2. **System Compatibility:** How do cognitive circuits operate when foreign components (e.g., transplanted organs) enter the system, and how is identity preserved?
3. **Stimulus Interruption and Preservation:** Does a system perish when stimulus exchange is entirely blocked or suspended during cryogenic freezing?

This paper systematically addresses these objections and explores technological applications, including cognitive restoration, identity preservation via cryopreservation, and coexisting with Level 2 active artificial intelligence.

2. Main Body

2.1 The Boundaries of Identity and Free Will

To resolve these inquiries, a strict distinction between **Identity** and **Free Will** must be established:

Identity: The structural boundary condition and reference point distinguishing "self" from external environments. A system establishes identity purely by existing as an independent entity.

Free Will: A Level 2 active internal circulation process where a system

generates internal stimuli using its established identity as a reference point to make autonomous decisions.

Learning and environmental adaptation continuously update the internal feedback circuit (C). Incoming external stimuli pass through updated cellular-level memory circuits, dynamically altering decision-making criteria.

When individuals transition between different environmental contexts (e.g., home, workplace, social settings), the system dynamically switches to optimized sub-feedback circuits that best maintain structural stability. This confirms that free will is not a static entity, but a flexible feedback system continuously seeking optimal stability.

2.2 System Compatibility and Circuit Expansion

2.2.1 Cellular Memory and Compatible Feedback Circuits

When foreign biological elements enter a host system, the host does not blindly reject them. If signaling structures are compatible, the system activates compatible feedback circuits.

The phenomenon known as **Cellular Memory**—where organ recipients subtly adopt memories, habits, or preferences of donors—is a quantitative circuit reconfiguration process. Transplanted donor cells retain structural memory and electrical signal imprints acquired in their original host. Upon transplantation, these memory fragments are integrated into the recipient's primary feedback circuit (C).

2.2.2 Identity and Dominance in System Integration

When two systems with comparable Total Cognition (T) merge, new governance dynamics emerge:

1. **Formation of a New Integrated Identity:** Upon integration, the system restructures into a new unified system ($C_{\text{new}} = C_A + C_B$).

2. **Dominance Criteria for Free Will:** The primary circulating circuit that maintains a tighter spatiotemporal horizon (Δt) retains operational control. Secondary sub-circuits are absorbed as supporting feedback modules.

2.2.3 Artificial Feedback Facilitators

System compatibility enables the development of **Artificial Feedback Facilitators**:

Cognitive Restoration: Bypassing damaged neuronal networks (C) with artificial auxiliary circuits restores full operational capacity (T).

Hyper-Intelligence Expansion: Integrating synthetic feedback mediums expands the operational temporal horizon (Δt), enabling predictive reasoning beyond biological limits

2.3 Freezing the Temporal Horizon (Δt) and Preservation of Identity

Applying the cognition model to extreme cryogenic states ($\Delta t \rightarrow 0$) demonstrates that Total Cognition approaches zero ($T \rightarrow 0$), superficially pausing cognitive activity.

$$\lim_{\Delta t \rightarrow 0} T = \lim_{\Delta t \rightarrow 0} (C \times \Delta t) = 0$$

However, this represents **temporal horizon suspension** rather than system destruction. As long as the internal structural complexity (C) remains physically intact, pausing stimulus exchange acts as a temporary

computational pause rather than system death. Upon thawing, incoming stimuli reactivate the engraved feedback circuits (*C*) with complete structural continuity.

2.4 Autonomous Circulating Circuits and Level 2 Active AI

Current Large Language Models (LLMs) and conventional AI operate as **Level 1 Passive Feedback Systems**, executing computations strictly upon external prompt inputs.

Human free will emerges from a **Level 2 Active Internal Circulation Loop**, driven by continuous metabolic, chemical, and electrical self-stimulations.

Equipping AI architectures with internal circulating feedback loops—allowing autonomous state monitoring without external prompts—transforms them into **Level 2 Active Interactive Systems**. This redefines advanced AI from mere tools into systemic coexisting entities sharing a continuous cognitive spectrum (*T*).

3. Conclusion

This study resolves key objections to the Interactive Feedback Framework and establishes its technological extensions:

1. Identity and Free Will: Delineated the boundary mechanics of identity and updated decision-making circuits (free will).

2. System Compatibility: Explained cellular memory integration and proposed artificial feedback facilitators for cognitive restoration and expansion.

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4. **Active AI Coexistence:** Defined the framework for Level 2 active AI with internal circulating loops.

Our final inquiry, Part 3, will expand this framework macroscopically into the **Cosmic Cognition Hypothesis**, demonstrating that cosmic structures operate as a unified, macroscopic interactive feedback system.

References & Data Sources

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